Multiclass Kidney Disease Risk Prediction Using Supervised Machine Learning Models

Machine Learning Final Project Report

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Course: Data Mining and Machine Learning

Abstract
Chronic kidney disease (CKD) is a major health concern because it can progress silently and may not be detected until kidney function has already declined. This project applied supervised machine learning models to classify kidney disease risk levels using a clinical and demographic dataset. The target variable represented five classes: High Risk, Low Risk, Moderate Risk, No Disease, and Severe Disease. The dataset contained 20,538 patient records and 43 original features, including laboratory indicators, symptoms, comorbidities, and lifestyle-related variables. The analysis involved data inspection, categorical encoding, class balancing using SMOTE, correlation analysis, feature engineering, model training, model evaluation, and deployment using Streamlit. Six machine learning models were implemented: Random Forest, XGBoost, K-Nearest Neighbors, Decision Tree, AdaBoost, and Voting Classifier. Random Forest achieved the best overall accuracy of 94.19%, followed by Voting Classifier at 92.30% and XGBoost at 88.35%. These results indicate that ensemble-based algorithms are highly effective for complex multiclass medical classification tasks.

1. Introduction

Chronic kidney disease is a long-term condition characterized by gradual loss of kidney function. Early identification of CKD risk is important because timely clinical intervention can reduce disease progression and improve patient outcomes. In real clinical settings, kidney disease assessment depends on multiple laboratory and clinical measurements, such as serum creatinine, blood urea, eGFR, urine characteristics, blood pressure, diabetes status, and hypertension history. Machine learning can help analyze these multivariable patterns and support automated risk classification.

The main research question of this project is: Can supervised machine learning models accurately classify chronic kidney disease risk levels using patient clinical and demographic data? This question guided the full workflow, starting from data inspection and preprocessing, then model development, performance evaluation, and final deployment. The project also aimed to compare different supervised learning models and identify which algorithm provides the best predictive performance for this medical dataset.

2. Dataset Description

The dataset used in this project contains 20,538 records and 43 columns before feature engineering. The features include numerical laboratory values, binary clinical indicators, categorical health-related variables, and demographic information. Examples of the clinical variables include blood pressure, specific gravity of urine, albumin in urine, sugar in urine, random blood glucose, blood urea, serum creatinine, sodium level, potassium level, hemoglobin level, eGFR, serum albumin, cholesterol, C-reactive protein, interleukin-6, diabetes duration, hypertension duration, and physical activity level.

The target column represents kidney disease risk category. The classes were encoded numerically as five categories: 0 = High Risk, 1 = Low Risk, 2 = Moderate Risk, 3 = No Disease, and 4 = Severe Disease. Initial data inspection showed that the dataset contained no missing values across the included columns, which allowed the project to proceed directly to encoding, resampling, and modeling steps.

Dataset Summary

Item	Description
Number of records	20,538 patient records
Original number of features	43 columns
Target type	Multiclass classification
Target classes	High Risk, Low Risk, Moderate Risk, No Disease, Severe Disease
Missing values	No missing values detected
Data types	Numerical and categorical clinical variables

3. Data Preprocessing

Preprocessing was necessary to convert the raw clinical dataset into a format suitable for machine learning. First, the dataset was inspected using basic information, descriptive statistics, and missing-value checks. Although no missing values were found, the dataset contained multiple categorical features that could not be directly used by most machine learning algorithms. Therefore, label encoding was applied to convert non-numeric variables into numerical values while preserving category distinctions.

Examples of encoded categorical variables include red blood cells in urine, pus cells in urine, pus cell clumps, bacteria in urine, hypertension, diabetes mellitus, coronary artery disease, appetite, pedal edema, anemia, family history of CKD, smoking status, physical activity level, and urinary sediment microscopy results. Encoding allowed these variables to be integrated into the modeling pipeline.

Class Imbalance Handling

The original target distribution was imbalanced, with the No Disease and Low Risk categories containing more observations than some disease-risk classes. This imbalance could bias a model toward majority classes and reduce its ability to detect minority risk groups. To address this issue, SMOTE was applied. After SMOTE, each class was balanced to 16,432 samples, creating a more equal training distribution across all five target classes.

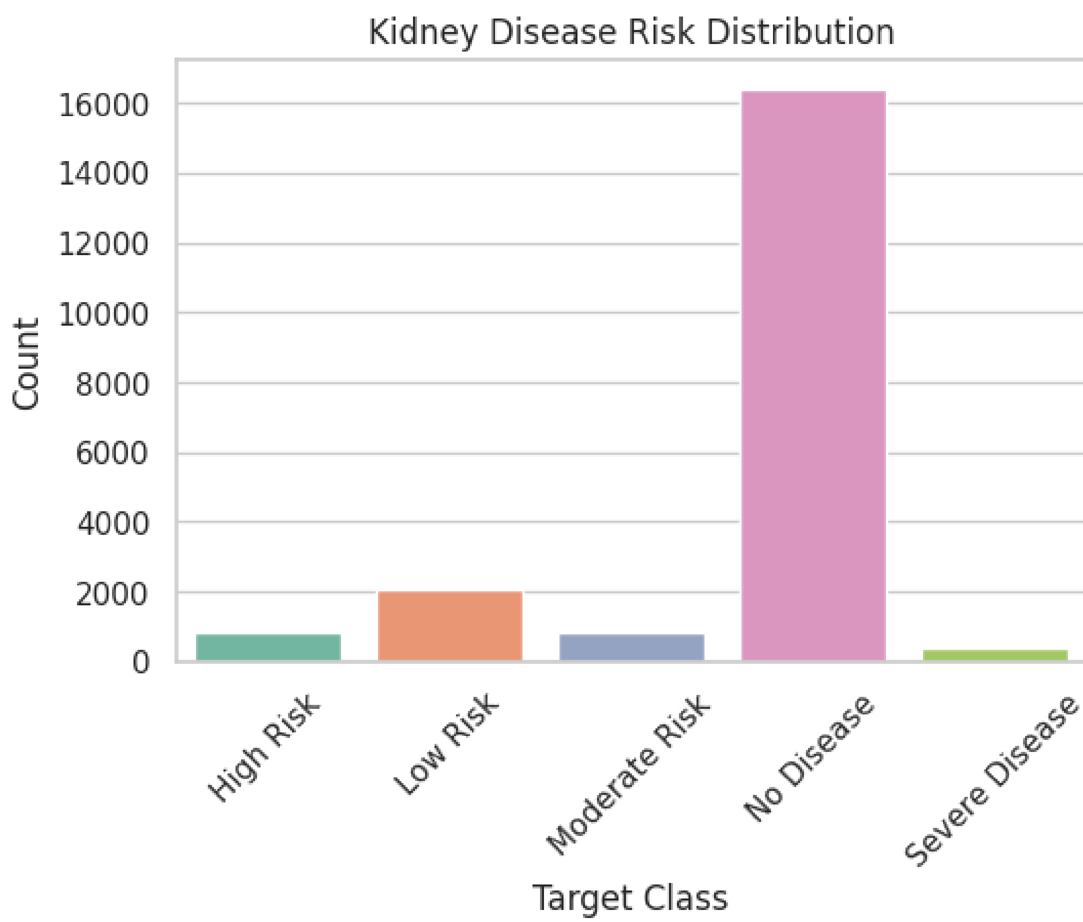

Figure 1. Target class distribution before balancing.

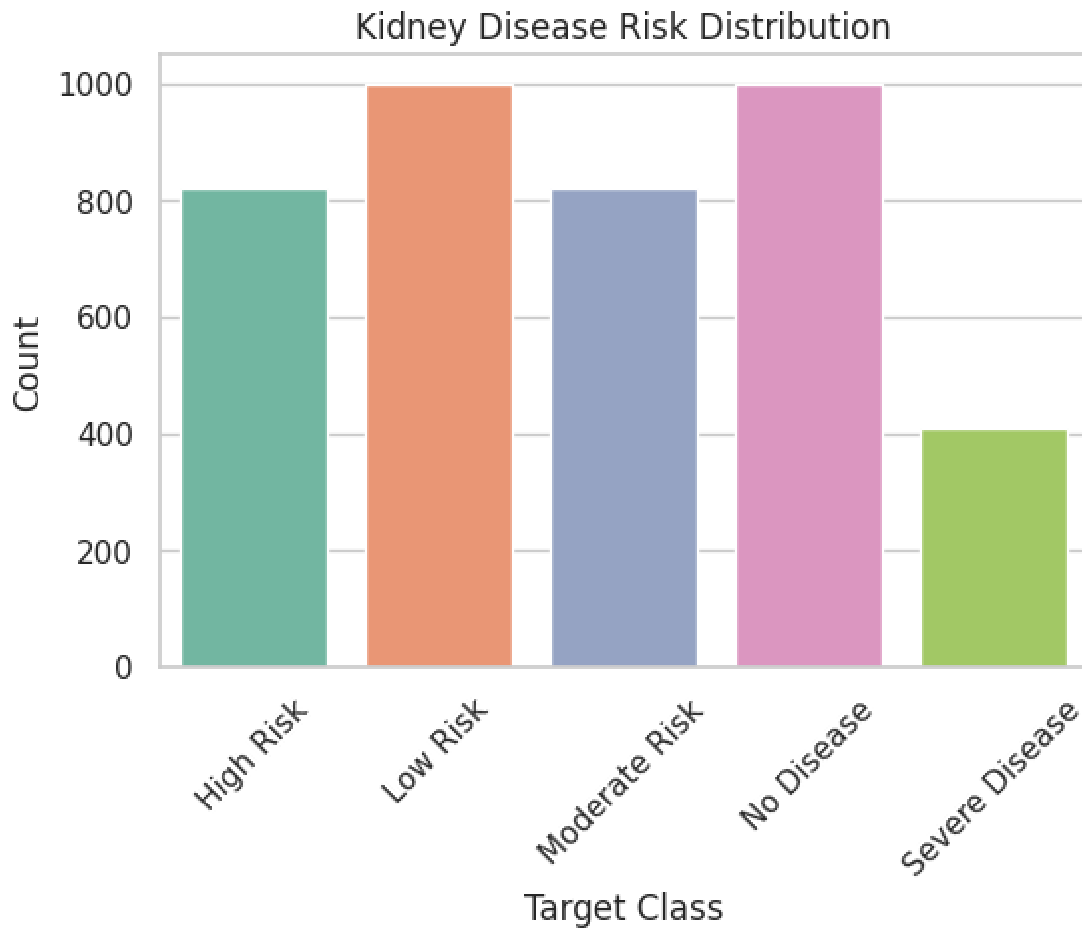

Figure 2. Target class distribution after manual resampling for visualization.

4. Feature Engineering and Feature Selection

Feature engineering was performed to create additional clinically meaningful variables from the existing patient data. These features were designed to represent relationships that may be relevant to kidney disease risk and improve model interpretability.

- Blood pressure to age ratio: calculated by dividing blood pressure by age plus one. This captures whether blood pressure is relatively high compared with patient age.
- Total disease duration: calculated by adding the duration of diabetes mellitus and duration of hypertension. This reflects cumulative exposure to two important CKD-related chronic conditions.
- Is elderly: a binary variable where patients above 60 years old were assigned 1 and younger patients were assigned 0.

Correlation analysis was also used as a filter-style feature selection method to examine how features relate to the target variable. Although many individual correlations were weak, some variables showed relatively higher associations with the target after balancing and feature engineering, including pus cells in urine, coronary artery disease, red blood cells in urine, pedal edema, hypertension, and family history of chronic kidney disease.

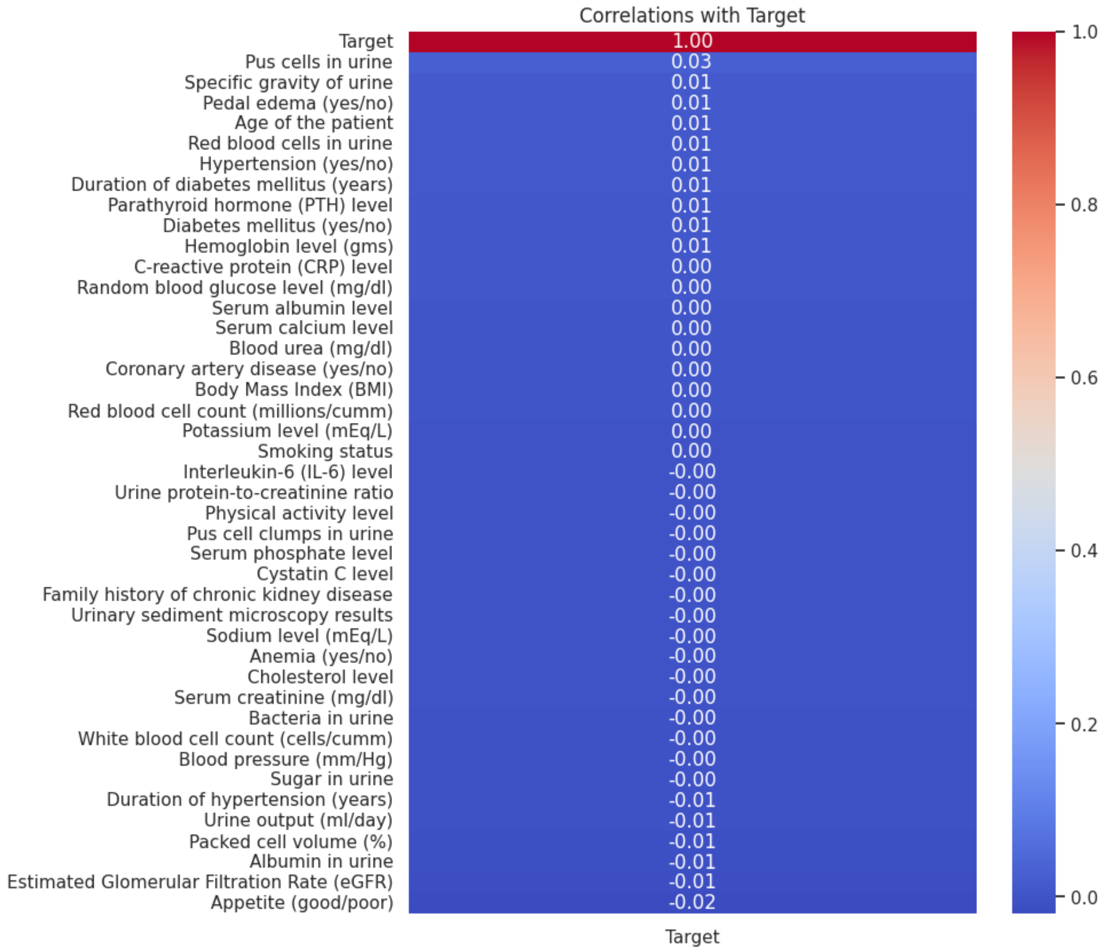

Figure 3. Correlation heatmap before resampling.

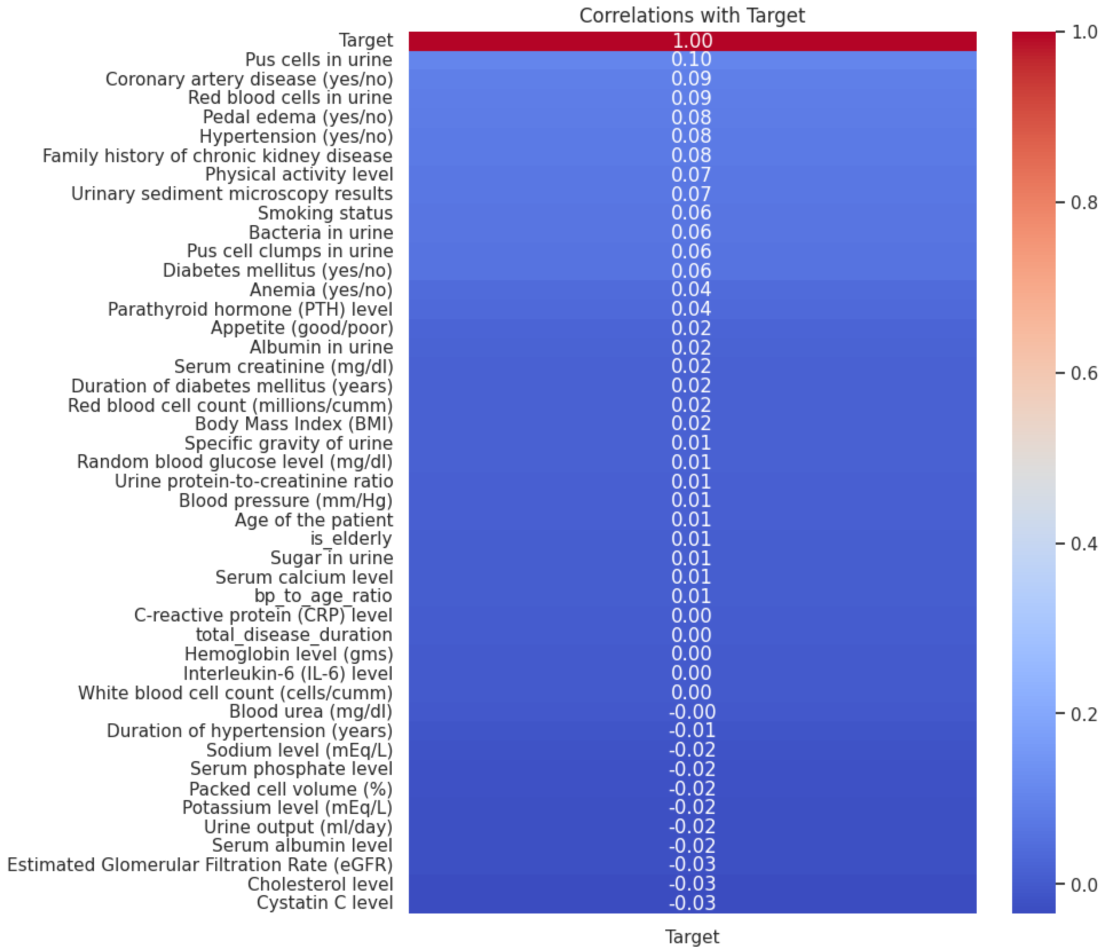

Figure 4. Correlation heatmap after SMOTE and feature engineering.

5. Model Selection and Training

After preprocessing and feature engineering, the dataset was split into training and testing subsets. Six supervised classification models were trained and evaluated to compare performance and identify the most suitable model for CKD risk prediction. The selected algorithms included both simple interpretable models and stronger ensemble methods.

- Random Forest: an ensemble method based on multiple decision trees and bagging. It is robust and performs well on complex tabular datasets.
- XGBoost: a gradient boosting algorithm known for high predictive accuracy and strong handling of nonlinear feature interactions.
- K-Nearest Neighbors: a distance-based classifier that predicts classes based on the closest training samples.
- Decision Tree: an interpretable rule-based model that can capture nonlinear relationships but may overfit.
- AdaBoost: a boosting approach that combines weak learners sequentially, but it may struggle with noisy or overlapping classes.
- Voting Classifier: an ensemble that combines predictions from multiple models using soft voting to improve robustness.

Performance was evaluated using accuracy, confusion matrices, classification reports, F1-score, and ROC-AUC. These metrics allowed assessment of both overall performance and class-wise predictive ability.

6. Results and Discussion

6.1 Random Forest Performance

Random Forest achieved the highest overall accuracy at 94.19%. The classification report showed strong precision, recall, and F1-score across all five classes. Class 4, representing Severe Disease, achieved excellent performance with precision of 1.00 and recall of 0.98. Class 3, No Disease, had slightly lower precision of 0.83 but strong recall of 0.92, suggesting that some other classes were predicted as No Disease but the model still detected most actual No Disease samples. Overall, Random Forest was the strongest individual model and was selected as the best-performing algorithm.

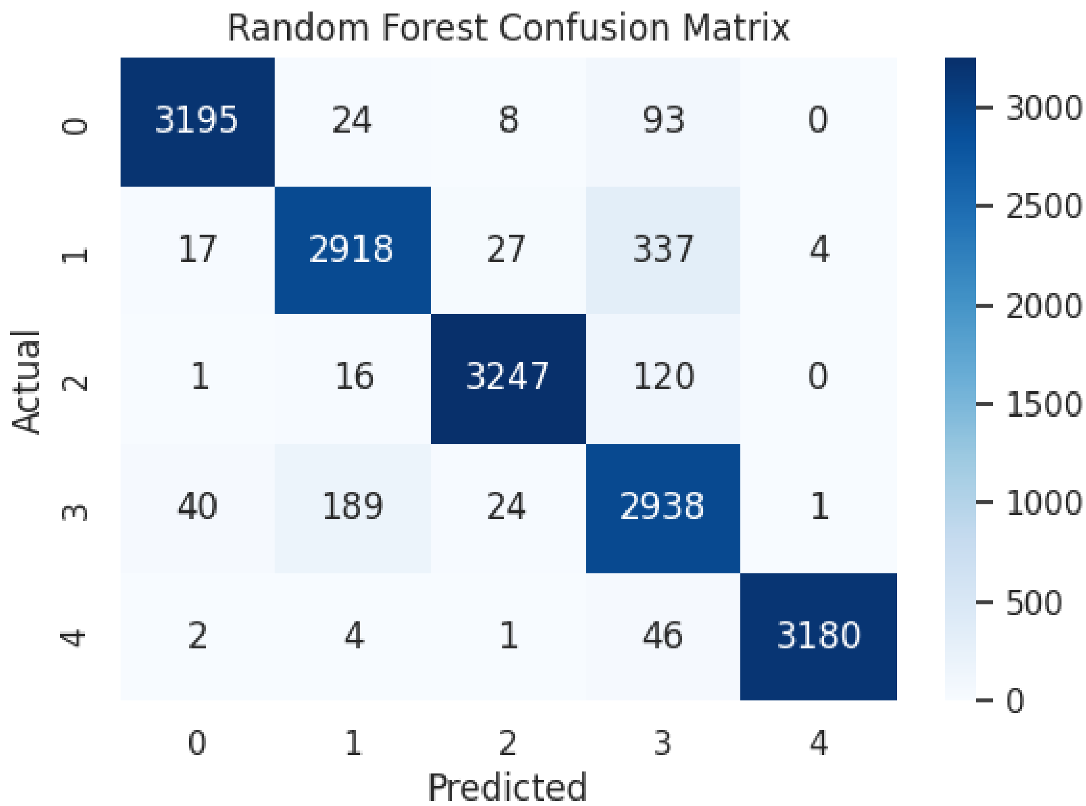

Figure 5. Random Forest confusion matrix.

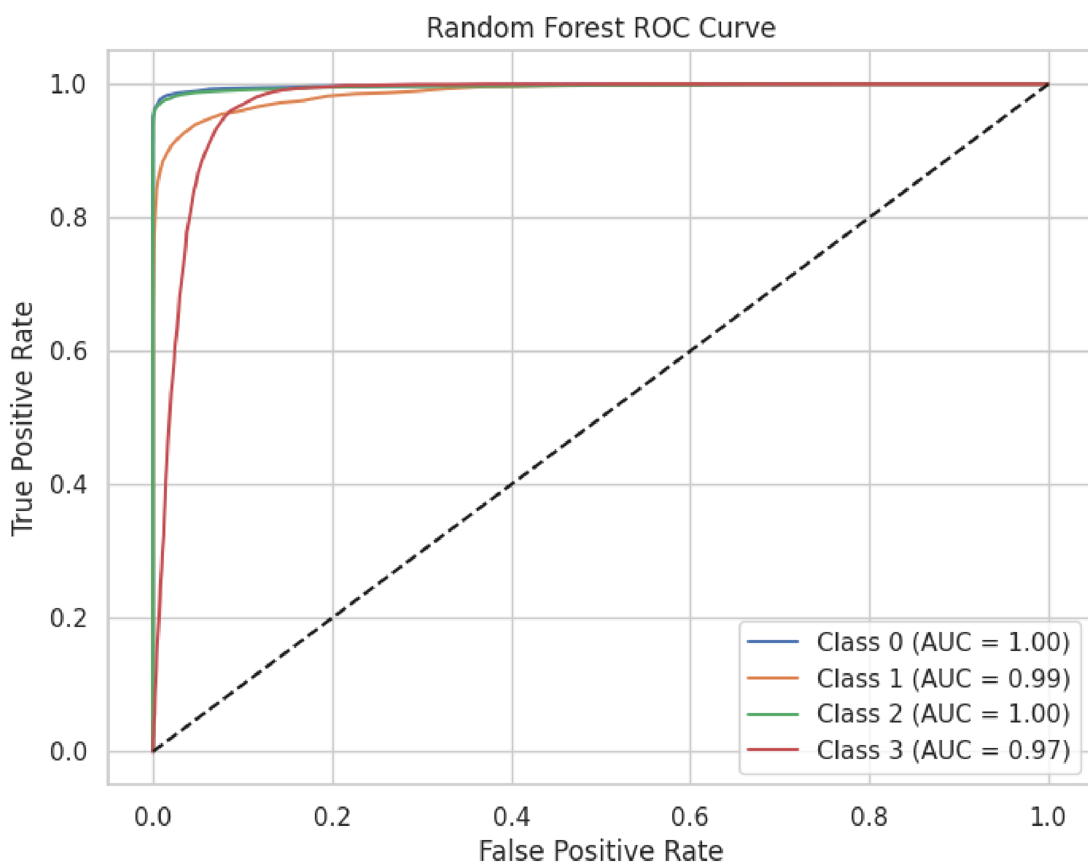

Figure 6. Random Forest ROC curve.

6.2 XGBoost Performance

XGBoost achieved an accuracy of 88.30%, which indicates strong performance but lower than Random Forest and Voting Classifier. The model performed particularly well for Severe Disease and No Disease classes, but recall for Class 1 was lower at 0.74. This suggests some difficulty in separating Low Risk cases from neighboring categories. Despite this limitation, the model showed strong multiclass discrimination with a macro F1-score of approximately 0.88 and AUC of 0.9766.

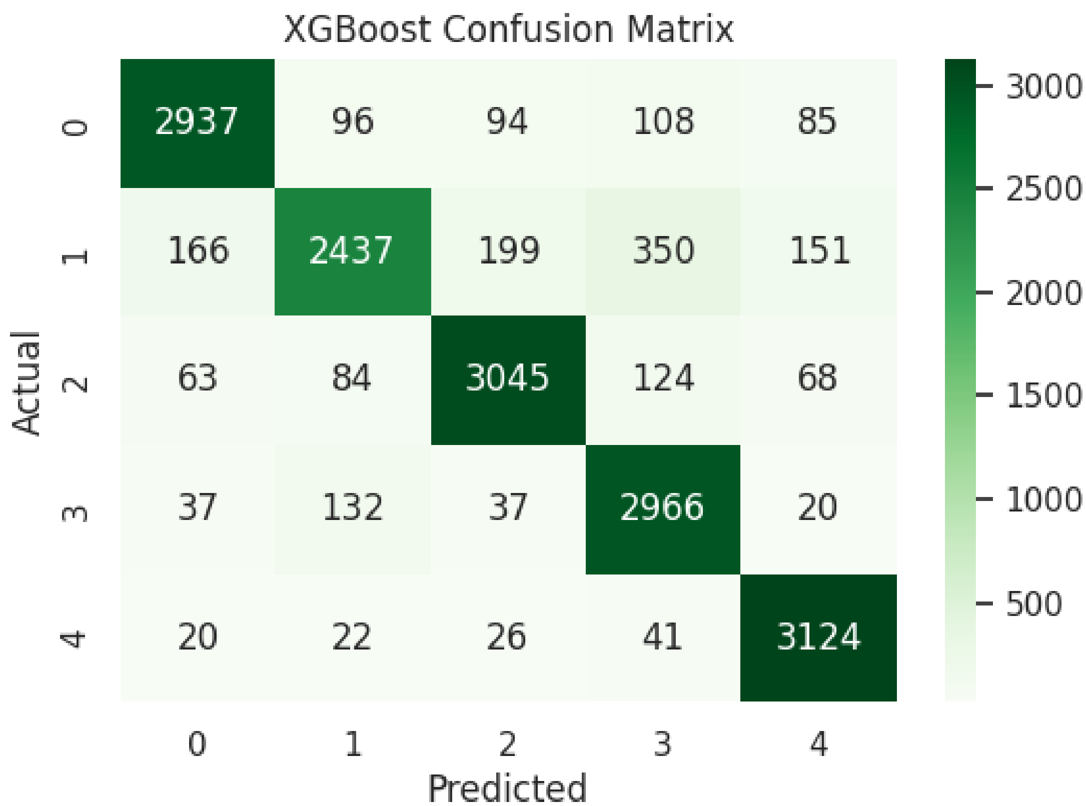

Figure 7. XGBoost confusion matrix.

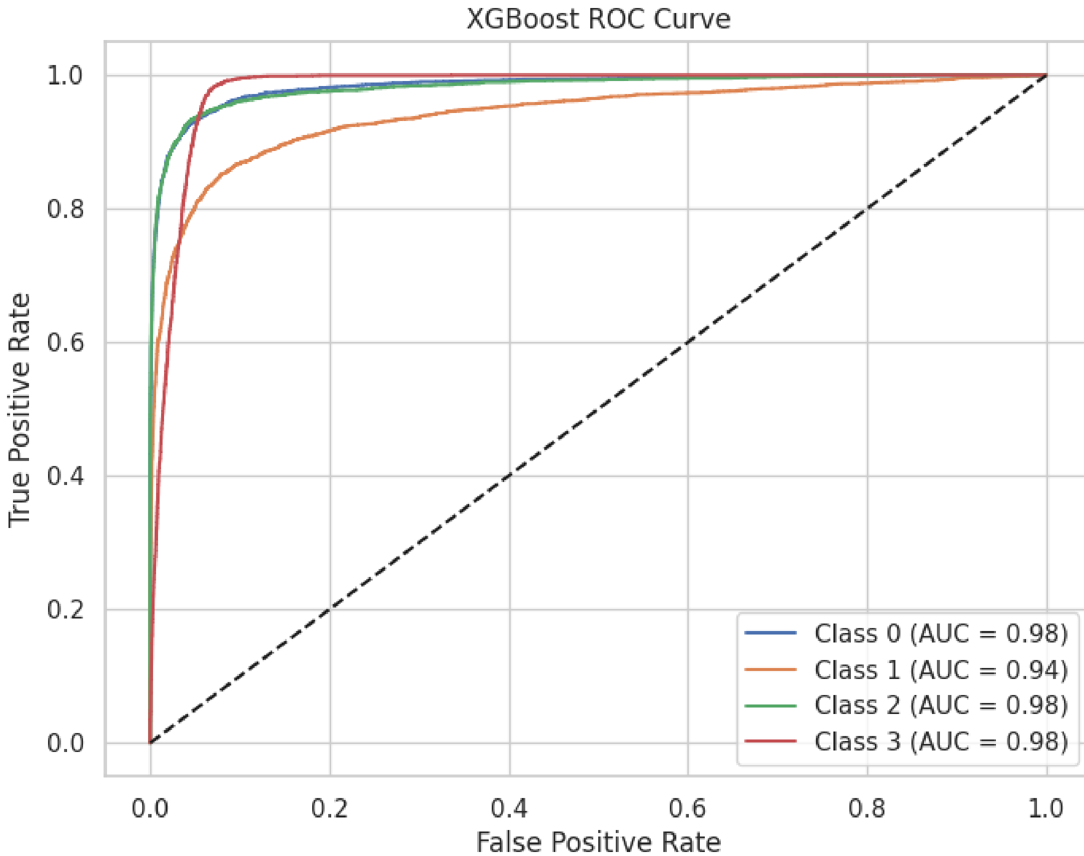

Figure 8. XGBoost ROC curve.

6.3 K-Nearest Neighbors Performance

KNN achieved an accuracy of 82.91%. The model predicted Classes 0, 1, 2, and 4 relatively well, but struggled with Class 3, where recall dropped to 0.26. This indicates that KNN had difficulty distinguishing No Disease cases from other risk groups, likely because distance-based classification is sensitive to overlapping feature patterns in high-dimensional medical data.

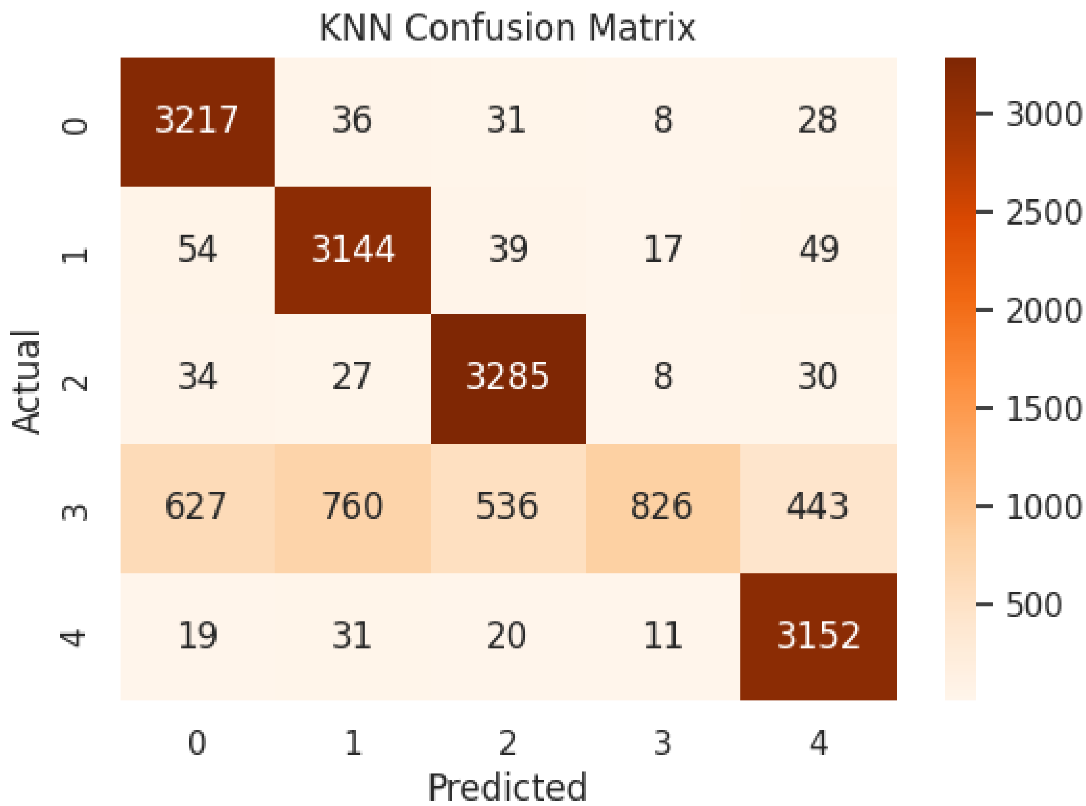

Figure 9. KNN confusion matrix.

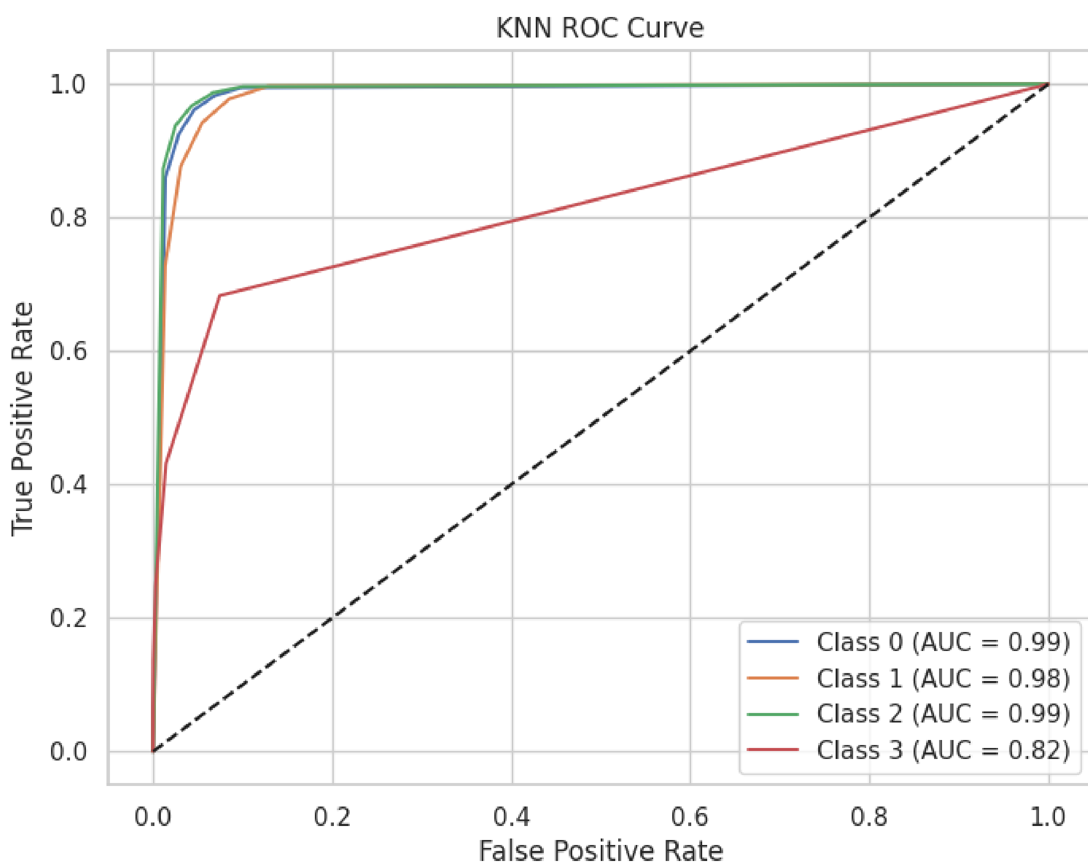

Figure 10. KNN ROC curve.

6.4 Decision Tree Performance

The Decision Tree model achieved 75.38% accuracy. Although it is easy to interpret, its performance was weaker than ensemble models. The confusion matrix showed more misclassification between Low Risk, Moderate Risk, and No Disease categories. This result suggests that a single decision tree may not capture the complexity of the relationships in the dataset as effectively as ensemble models.

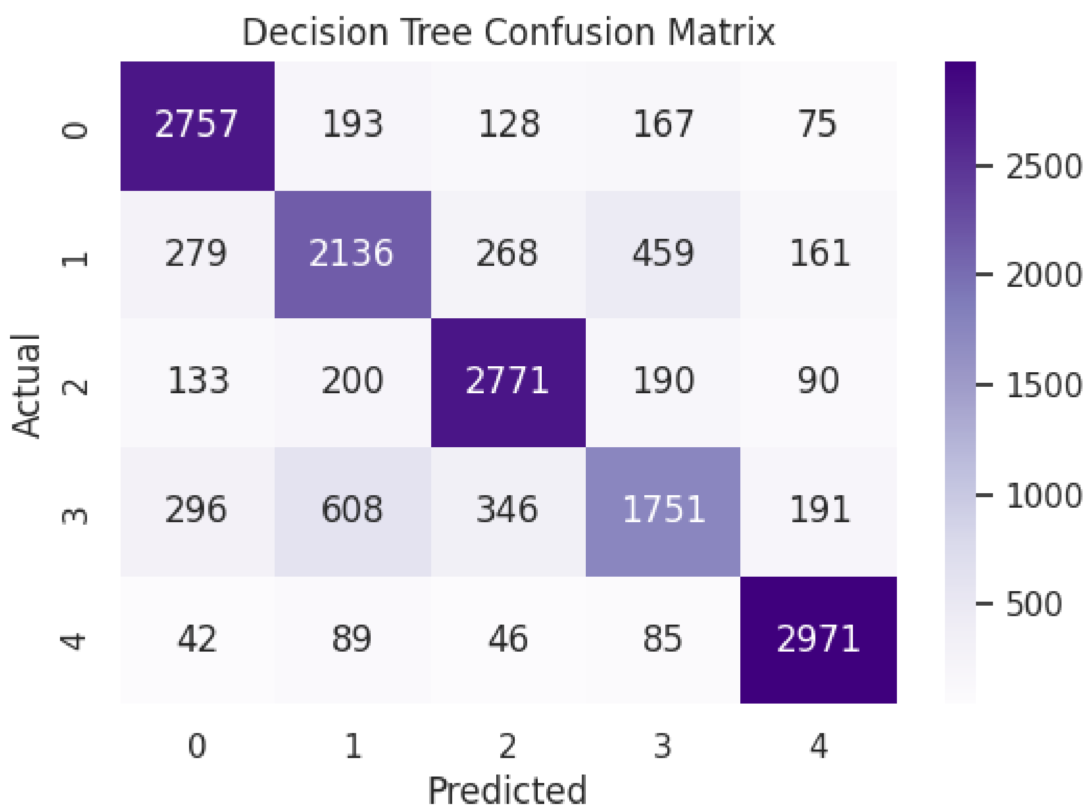

Figure 11. Decision Tree confusion matrix.

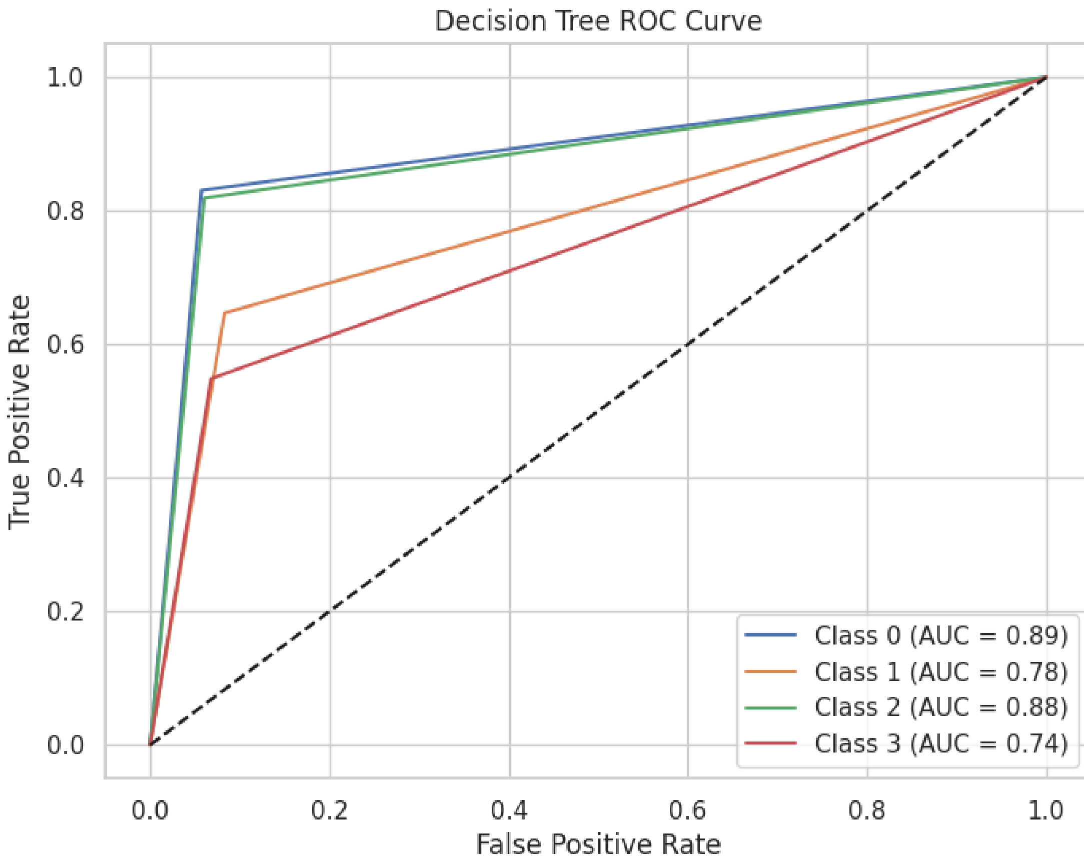

Figure 12. Decision Tree ROC curve.

6.5 AdaBoost Performance

AdaBoost had the weakest performance, with an accuracy of 34.43%. The model showed widespread misclassification across multiple classes and had low macro-averaged F1-score. Although AdaBoost can be powerful in some tasks, this result suggests that the selected weak learners were not sufficient to model the complex and overlapping structure of this multiclass medical dataset.

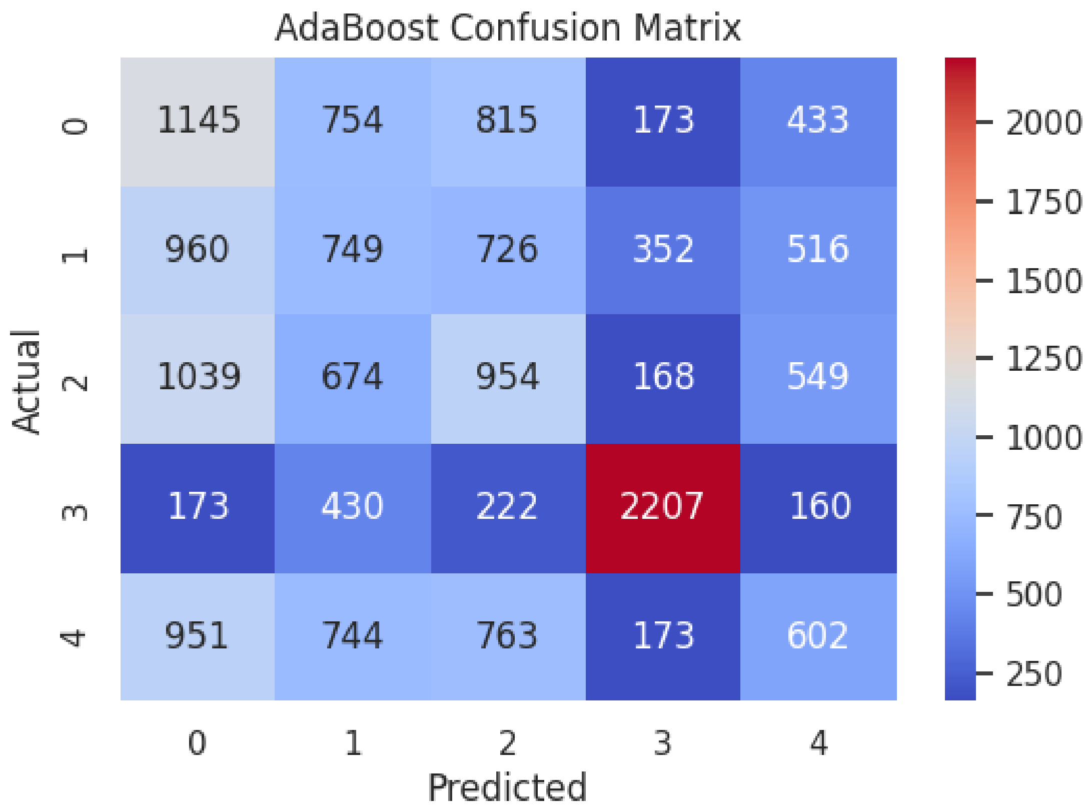

Figure 13. AdaBoost confusion matrix.

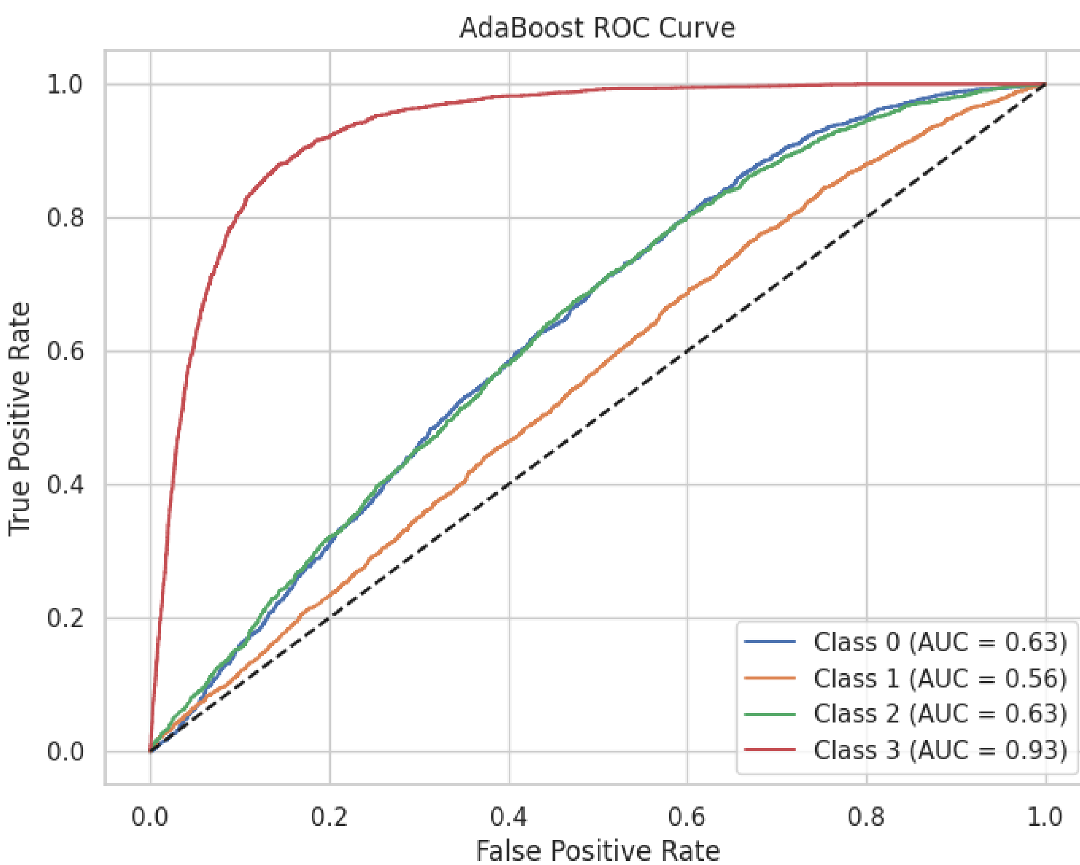

Figure 14. AdaBoost ROC curve.

6.6 Voting Classifier Performance

The Voting Classifier achieved 92.30% accuracy, making it the second-best model after Random Forest. It combined multiple base learners using soft voting and produced strong, balanced performance across the five target classes. The model achieved high precision and recall for High Risk, Moderate Risk, and Severe Disease classes, although No Disease had slightly lower recall at 0.80. The strong AUC of 0.9925 indicates excellent overall class separability.

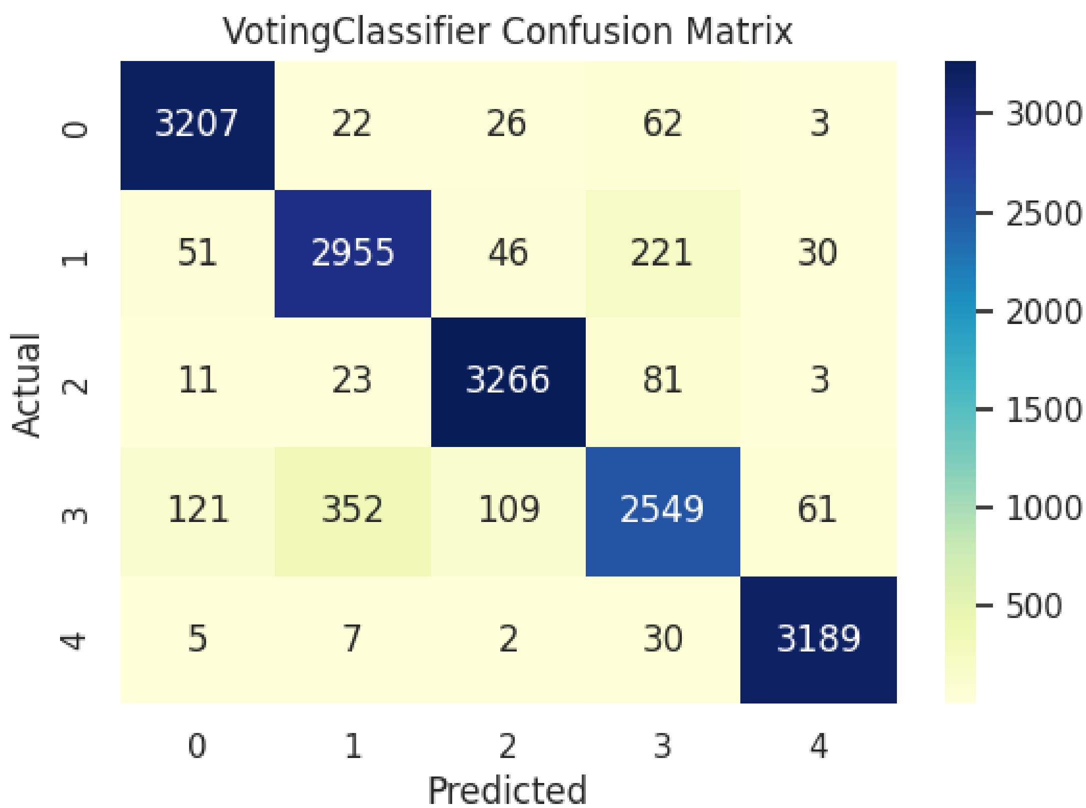

Figure 15. Voting Classifier confusion matrix.

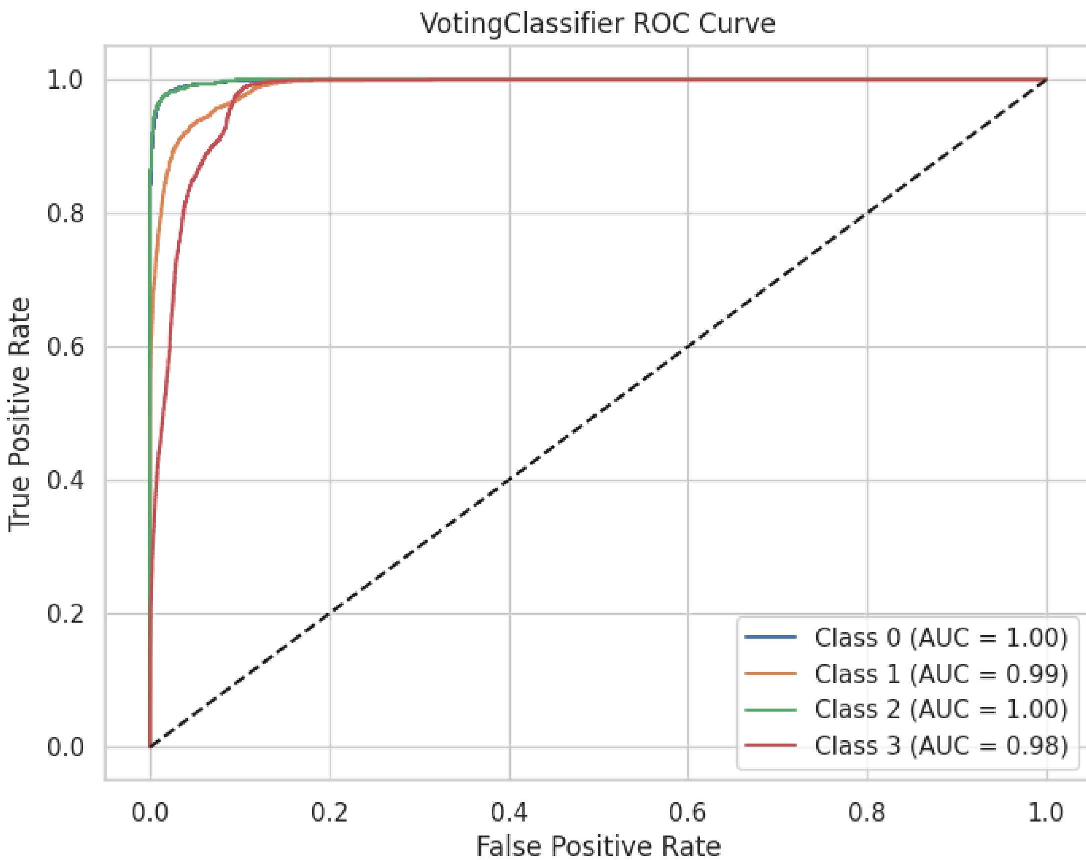

Figure 16. Voting Classifier ROC curve.

7. Model Comparison

The overall accuracy comparison shows that ensemble-based models performed best. Random Forest achieved the highest accuracy at 94.19%, followed closely by Voting Classifier at approximately 92.30%. XGBoost also produced strong results with 88.30% accuracy. KNN and Decision Tree showed moderate performance, while AdaBoost performed poorly. The F1-score and AUC comparison further confirms that Random Forest and Voting Classifier were the strongest models for this classification task.

Model	Accuracy	Macro F1	AUC	Interpretation
Random Forest	94.19%	0.9423	0.9909	Best individual model
Voting Classifier	92.30%	0.9215	0.9925	Strong ensemble model
XGBoost	88.30%	0.8815	0.9766	Good performance
KNN	82.91%	0.7925	0.9529	Weak on No Disease recall
Decision Tree	75.38%	0.7487	0.8456	Interpretable but weaker
AdaBoost	34.43%	0.3457	0.6757	Weakest model

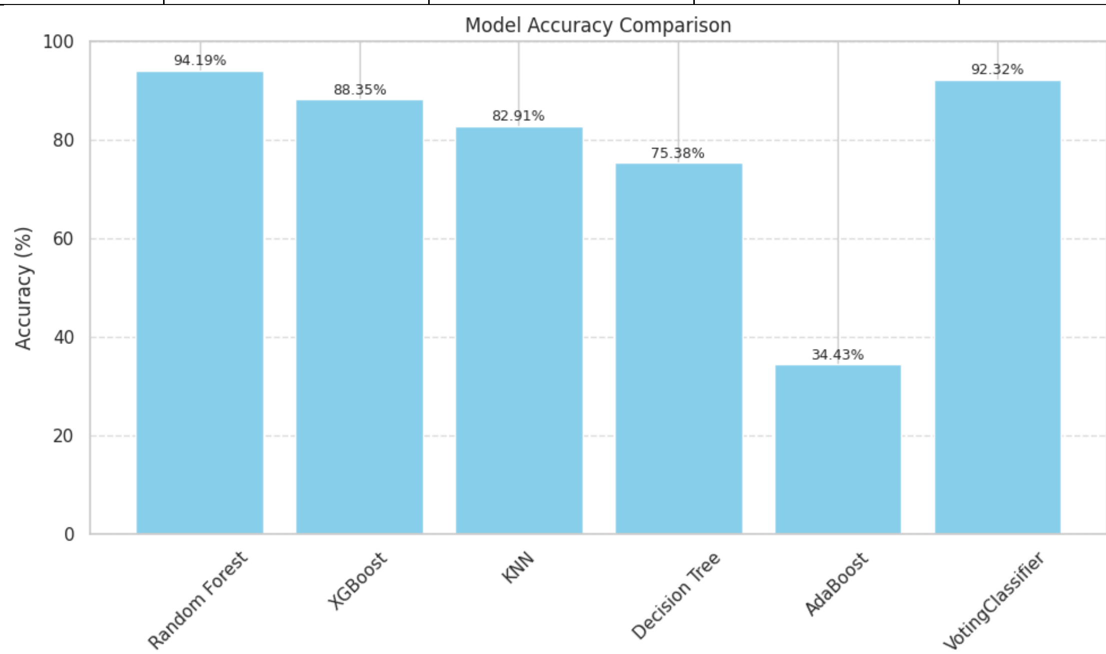

Figure 17. Model accuracy comparison.

8. Deployment

The project was deployed using the Streamlit framework. The purpose of deployment was to provide a web application link that demonstrates the project workflow and presents the machine learning solution in an accessible format. The GitHub repository contains the notebook, dataset, app.py file, requirements.txt file, and saved preprocessing file. Due to model file size limitations during GitHub upload, the final deployed Streamlit version was simplified to display the project title, workflow, best model, and conclusion, while the full training and evaluation process remains documented inside the notebook.

Deployment link: https://ckd-ml-project-kvvbcivm5si5jcu7vmqwz.streamlit.app/

9. Limitations

Although the project achieved strong predictive performance, some limitations should be considered. First, the dataset is synthetic or compiled and may not fully represent real clinical variability across hospitals or populations. Second, SMOTE was useful for balancing classes, but synthetic oversampling may introduce artificial patterns that do not perfectly match real patients. Third, some classes showed overlap, especially Low Risk and No Disease, which caused misclassification in several models. Fourth, the deployed Streamlit app was simplified due to large model file limitations, so the live prediction interface was not fully included in the final web version.

10. Conclusion

This project successfully demonstrated the use of supervised machine learning for multiclass chronic kidney disease risk prediction. The dataset was inspected, encoded, balanced using SMOTE, enhanced through feature engineering, and used to train several classification models. Random Forest produced the best overall performance with 94.19% accuracy, strong F1-scores, and high AUC values. The Voting Classifier also performed strongly, confirming the value of ensemble learning methods for structured clinical data. Overall, the results support the research question by showing that machine learning can effectively classify CKD risk levels using clinical and demographic patient information.

Video link